

A Variational Graph Neural Network with Multi-view Integration for Predicting LncRNA–Disease Associations

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Abstract. Long non-coding RNAs (lncRNAs) are critical regulators in diverse biological processes, and their dysregulation is implicated in numerous diseases. Identifying lncRNA-disease associations is crucial for understanding disease mechanisms and discovering therapeutic targets. However, experimental validation is labor-intensive and time-consuming, while existing computational methods struggle with network sparsity and inadequate integration of heterogeneous biological data. We propose a novel end-to-end deep learning framework that integrates attention-weighted multi-view graph convolutional networks (GCNs), a variational graph autoencoder (VGAE), and a gated multilayer perceptron to predict potential lncRNA-disease associations. The method constructs a unified heterogeneous graph from multiple similarity networks of lncRNAs, diseases, and miRNAs, along with known associations. This architecture first employs attention-weighted GCNs to extract deep topological features, then uses a VGAE to learn robust, probabilistic topology-aware node embeddings, and finally predicts associations through a specialized gated MLP predictor. Comprehensive evaluation on a primary benchmark dataset 1 demonstrates good performance (AUC 0.9584, AUPR 0.9590), with a second dataset confirming generalizability, proving the approach effectively addresses data sparsity, heterogeneity, and class imbalance as a robust tool for prioritizing lncRNA-disease candidates.

Keywords: lncRNA-disease associations · Variational Graph Autoencoder · Graph neural networks · Multi-view learning

1 Introduction

Long non-coding RNAs (lncRNAs) are transcripts longer than 200 nucleotides that do not code for proteins but regulate gene expression at multiple levels. Their dysregulation is implicated in diverse diseases (e.g., cancer-related HOTAIR [22] and context-dependent MALAT1 [23]), making them promising

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biomarkers and therapeutic targets. However, most lncRNAs remain functionally uncharacterized and experimental validation is costly, motivating computational discovery of lncRNA–disease associations (LDAs). Existing computational approaches broadly fall into matrix factorization, network propagation, and machine learning/deep learning families.

Existing methods for predicting LDAs include matrix factorization (MF), network propagation, and learning-based models. MF approaches learn latent representations from incomplete association matrices, e.g., NRMFLDA [6], but they can inherit noise, require careful hyperparameter tuning, and degrade under cold-start or positive–unlabeled settings. Network propagation methods diffuse evidence on similarity graphs; OS–LDA [19] exemplifies this paradigm and often yields interpretable paths or neighbors, yet diffusion can be sensitive to graph noise, biased toward hubs, and fragile for sparse or isolated nodes. Learning-based methods leverage heterogeneous biomedical data via feature engineering or representation learning: classical ML models such as LDA-GARB [16] use handcrafted descriptors, while deep architectures such as GCNFORMER [20] and graph autoencoders (GEnDDn [14], LDAGM [21]) learn nonlinear embeddings from multi-view graphs. Despite strong predictive power, learning-based methods still face noise sensitivity, overfitting with scarce positives, distribution shifts for unseen entities, and limited interpretability, motivating models that integrate multi-modal signals while remaining robust under sparsity.

To address these limitations, we propose an end-to-end model for predicting lncRNA–disease associations by unifying attention-weighted multi-view GCNs, variational graph inference, and a deep link predictor. From heterogeneous similarity networks of lncRNAs, diseases, and miRNAs together with observed interactions, our method builds a unified heterogeneous graph, learns topology-aware node embeddings via a Variational Graph Autoencoder, and scores candidate links through a gated MLP classifier. A joint objective couples graph reconstruction with link classification and addresses class imbalance through a principled positive weighting scheme. Under five-fold evaluation, the model delivers strong and consistent performance across evaluation metrics, demonstrating robustness to sparsity, heterogeneity, and nonlinearity in the LDA task.

2 Materials and Methods

2.1 Datasets

In this study, to demonstrate the effectiveness of our proposed model, we evaluate it on two datasets:

– **Dataset1** Dataset 1 was generated by Zhou *et al.* [24]. It integrates associations for 665 lncRNAs, 316 diseases, and 295 miRNAs. The data includes 3,833 lncRNA-disease associations from Lnc2Cancer v3.0 [5] and LncRNADisease v2.0 [11], 2,108 lncRNA-miRNA associations from starBase v2.0 [9], and 8,540 miRNA-disease associations from HMDD v3.0 [7].

– **Dataset2** We also utilized a dataset introduced by Fu *et al.* [3], which is constructed from 240 lncRNAs, 412 diseases, and 495 miRNAs. This dataset aggregates 2,697 lncRNA-disease associations (sourced from LncRNADisease [1], Lnc2Cancer [5], and GeneRIF [12]), 2,004 lncRNA-miRNA associations (1,002 from startBase v2.0 [9] and 1,002 from HMDD v2.0 [10]), and 13,562 miRNA-disease associations (from HMDD v2.0 [10]).

We conducted a comprehensive benchmark comparison on Dataset 1 and utilized Dataset 2 to further validate the effectiveness and generalizability of our model.

2.2 Proposed Method

Our model operates in two phases: (1) we construct multiple biological similarity networks and pre-computing topological features; (2) these pre-computed features are combined with features learned dynamically by a joint end-to-end model to produce final association predictions.

2.2.1 Phase 1: Multi-view Network Construction and Non-linear Feature Pre-computation

This phase constructs multiple biological similarity networks and generates pre-computed non-linear features for lncRNAs, diseases, and miRNAs, which are then combined with dynamically learned features in the end-to-end model.

Multi-view Similarity Computation

– **Disease Semantic Similarity**
Disease semantic similarity is computed from the Disease Ontology (DO) [15] using the DAG-based contribution scheme of Wang *et al.* [18] Specifically, each disease d_i is represented by $DAG_{d_i} = (T_{d_i}, E_{d_i})$, and the contribution of an ancestor node d to d_i is

$$W_{d_i} = \begin{cases} 1, & \text{if } d_i = d, \\ \max\{\Delta \times W_{d_i}(d') \mid d' \in \text{children of } d\}, & \text{if } d_i \neq d, \end{cases} \tag{1}$$

where $\Delta = 0.5$ is the attenuation factor. The semantic similarity between diseases d_i and d_j is

$$DSim(d_i, d_j) = \frac{\sum_{d \in D_i \cap D_j} \left(W_{d_i}(d) + W_{d_j}(d)\right)}{DS(d_i) + DS(d_j)} \tag{2}$$

where $DS(d_i) = \sum_{d \in D_i} W_{d_i}(d)$. The resulting disease semantic similarity matrix is denoted **DSM**.

– **lncRNA and miRNA Functional Similarity**
Following Wang *et al.* [17] we compute lncRNA/miRNA functional similarity by matching the disease sets associated with R_i and R_j using $DSim(\cdot, \cdot)$:

$$FSim(R_i, R_j) = \frac{\sum_{i=1}^{n_1} \max_{1 \le j \le n_2} DSim(d_{1i}, d_{2j}) + \sum_{j=1}^{n_2} \max_{1 \le i \le n_1} DSim(d_{2j}, d_{1i})}{n_1 + n_2} \tag{3}$$

where n_1 and n_2 are the numbers of associated diseases for R_i and R_j. This yields the lncRNA and miRNA functional similarity matrices **LFM** and **MFM**.

– **Gaussian Interaction Profile (GIP) Kernel Similarity**
Following Chen *et al.* [2] we compute GIP kernel similarity from interaction profiles:

$$GSim(R_i, R_j) = \exp\left(-\gamma_l \|IP(R_i) - IP(R_j)\|^2\right), \tag{4}$$

where $IP(\cdot)$ denotes the interaction profile. The kernel width is normalized as

$$\gamma_l = \frac{\gamma_l'}{\frac{1}{n_l}\sum_{i=1}^{n_l} \|IP(R_i)\|^2}, \tag{5}$$

and similarly for miRNAs and diseases:

$$GSim_m(M_i, M_j) = \exp\left(-\gamma_m \|IP(M_i) - IP(M_j)\|^2\right), \tag{6}$$

$$GSim_d(d_i, d_j) = \exp\left(-\gamma_d \|IP(D_i) - IP(D_j)\|^2\right). \tag{7}$$

This yields the GIP similarity matrices **LGM**, **MGM**, and **DGM**.

Deep Feature Fusion and Enhancement

To mitigate biological data sparsity, we apply a two-step feature enhancement process before the main learning stages.

Step 1: Deep Topology Feature Extraction

Inspired by LDAGM [21], we iteratively refine similarity representations (Fig. 1). We first fuse similarity sources and normalize:

$$X \in \{L, M, D\}: \quad XM_1 = \frac{XFM_1 + XGM_1}{\max(XFM_1 + XGM_1)}. \tag{8}$$

For $k \ge 2$, we update each source by multiplying with the previous fused matrix, e.g., $LFM_k = LM_{k-1} \otimes LFM_{k-1}$ and $LGM_k = LM_{k-1} \otimes LGM_{k-1}$ (analogously for miRNAs and diseases), and then re-fuse:

$$LM_k = \frac{LFM_k + LGM_k}{\max(LFM_k + LGM_k)}. \tag{9}$$

This recursive process is repeated for all three entity types. After the final iteration, the fused similarity matrices (LM_i, DM_i, MM_i) are combined with

the interaction matrices to construct the multi-view heterogeneous adjacency matrix, A_i:

$$A_i = \begin{bmatrix} LM_i & LD & LM \\ DL & DM_i & DM \\ ML & MD & MM_i \end{bmatrix} \tag{10}$$

This comprehensive network serves as the input for the Graph Convolutional Autoencoder (GCN-AE) for nonlinear feature extraction.

Step 2: Graph Convolutional Autoencoder

As illustrated in Fig. 2, each similarity view is processed by a Graph Convolutional Autoencoder (GCN-AE), which consists of two primary components: a GCN-based encoder and a linear decoder.

- **GCN Encoder**
  The encoder takes the heterogeneous adjacency matrix A_i as input. To reduce noise in the input data, we first apply Laplace normalization:

$$L = D^{-\frac{1}{2}} A_i D^{-\frac{1}{2}} \tag{11}$$

  where D is the diagonal degree matrix of A_i. A graph convolution operation followed by a linear transformation produces the final embedding R_e:

$$R_e = \mathrm{ReLU}[(A_i \times L)W + b] \tag{12}$$

  where W is a learnable weight matrix, b is a learnable bias term, and ReLU is the nonlinear activation function.
- **Linear Decoder and Training**
  The decoder reconstructs the original adjacency matrix from the embedding R_e using a bilinear decoding layer:

$$R_d = [\mathrm{ReLU}(R_e W + b)]W + b \tag{13}$$

  The model is trained by minimizing the mean square error (MSE) between the decoder's output R_d and the original input matrix, A_i.

$$\mathrm{Loss} = \frac{1}{n} \sum_{i=1}^{n} (y_i - y_i')^2 \tag{14}$$

  This training process allows the encoder to learn informative feature embeddings representing the network's topology.

2.2.2 Phase 2: Integrated Prediction Model

The second phase of our model is an integrated model that learns task-specific features and combines them with the pre-computed features from Phase 1 to predict lncRNA-disease associations.

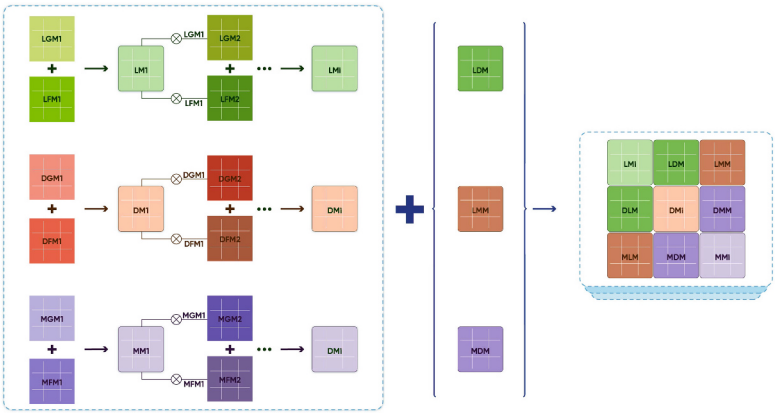


Fig. 1. Deep topological feature extraction and multi-view heterogeneous network construction. The process fuses multiple similarity measures (functional, semantic, and GIP) to generate topological features for lncRNAs, miRNAs, and diseases. These features are then integrated to build the comprehensive heterogeneous network.

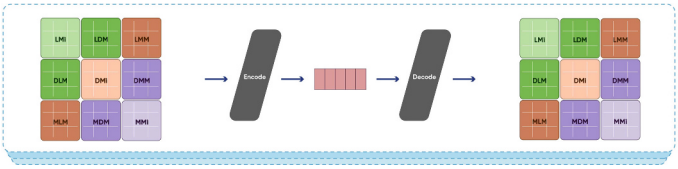


Fig. 2. Nonlinear feature extraction. The heterogeneous network is processed by a GCN encoder to generate embeddings. A bilinear decoder then reconstructs the network, achieving dimensionality reduction.

GCN and Attention-based Feature Fusion and Heterogeneous Network Reconstruction

To learn task-specific features, we treat each similarity matrix as a separate view (Fig. 3). Each view is encoded by a two-layer GCN, and the resulting embeddings are fused by node-wise self-attention to obtain a single representation that is used to reconstruct enriched similarity matrices and the final heterogeneous network.

A two-layer Graph Convolutional Network (GCN) learns node embeddings per view, producing view-specific representations $\{Z^{(1)}, Z^{(2)}, ..., Z^{(V)}\}$. The propagation rule is:

$$H^{(l+1)} = \sigma(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}) \tag{15}$$

where $\hat{A} = A + I$ adds self-loops, $\hat{D}$ is the degree matrix, and $H^{(0)}$ is the input feature matrix (identity in our implementation).

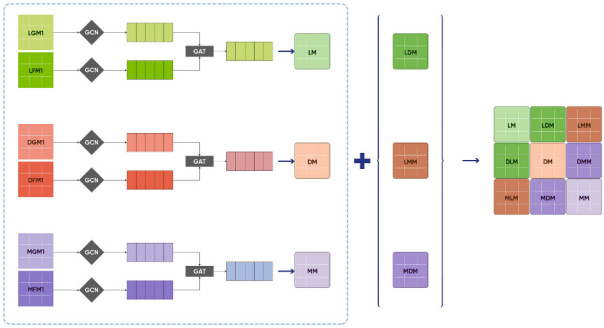


Fig. 3. Integrated model for GCN and attention-based feature fusion. Each similarity matrix is processed independently by a two-layer GCN to learn view-specific embeddings, which are then fused through a self-attention mechanism to form heterogeneous network.

To integrate multiple views, a self-attention layer learns node-wise view weights. The attention score e_{iv} is:

$$e_{iv} = w_a^T \tanh(W_a z_i^{(v)} + b_a) \tag{16}$$

Softmax normalization yields attention weights α_{iv}:

$$\alpha_{iv} = \frac{\exp(e_{iv})}{\sum_{k=1}^{V} \exp(e_{ik})} \tag{17}$$

The fused embedding z_i^{final} is a weighted sum of view-specific embeddings:

$$z_i^{final} = \sum_{v=1}^{V} \alpha_{iv} z_i^{(v)} \tag{18}$$

We obtain final embedding matrices for lncRNAs, diseases, and miRNAs, denoted $Z^{\text{final}}_{\text{lncRNA}}$, $Z^{\text{final}}_{\text{disease}}$, and $Z^{\text{final}}_{\text{miRNA}}$ by transforming the attention-weighted embeddings through a multi-layer perceptron. These are used to reconstruct enriched similarity matrices via a sigmoid-activated inner product, e.g.:

$$LM_{\text{enriched}} = \text{sigmoid}(Z^{\text{final}}_{\text{lncRNA}} (Z^{\text{final}}_{\text{lncRNA}})^T) \tag{19}$$

Similarly, we reconstruct DM_{enriched} and MM_{enriched}, and then assemble the final heterogeneous network A_{final} following Eq. (10):

$$A_{\text{final}} = \begin{bmatrix} LM_{\text{enriched}} & LD & LM \\ DL & DM_{\text{enriched}} & DM \\ ML & MD & MM_{\text{enriched}} \end{bmatrix} \tag{20}$$

The resulting A_{final} is passed to the subsequent topological learning stage.

Topological Representation Learning with a Variational Graph Autoencoder

We apply a Variational Graph Autoencoder (VGAE) to A_{final} to learn low-dimensional probabilistic node embeddings. As illustrated in Fig. 4, the VGAE uses a GCN encoder and an inner-product decoder.

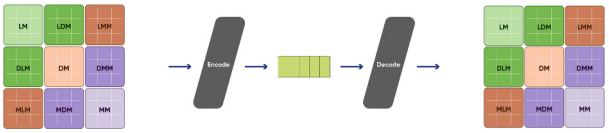


Fig. 4. Topological representation learning with a VGAE. The heterogeneous network is encoded into a probabilistic latent space via a GCN, which outputs mean and variance parameters. The decoder reconstructs the network from sampled embeddings, optimizing reconstruction and KL divergence losses.

The encoder outputs a mean μ and log-variance $\log \sigma^2$ per node, and samples embeddings using $z = \mu + \epsilon \odot \sigma$.

The decoder reconstructs the graph structure via an inner product:

$$\hat{A} = \text{sigmoid}(ZZ^T) \tag{21}$$

Training minimizes a reconstruction loss and a KL regularizer. Reconstruction uses a weighted binary cross-entropy (weight w emphasizes observed links, $A_{ij} = 1$):

$$\mathcal{L}_{\text{recon}} = -\frac{1}{N^2} \sum_{i,j} (w \cdot A_{ij} \log(\hat{A}_{ij}) + (1 - A_{ij}) \log(1 - \hat{A}_{ij})) \tag{22}$$

The KL term encourages a smooth latent space:

$$\mathcal{L}_{KL} = \frac{1}{2N} \sum_{i=1}^{N} \sum_{j=1}^{D_{\text{vgae}}} (1 + \log((\sigma_{ij})^2) - (\mu_{ij})^2 - (\sigma_{ij})^2) \tag{23}$$

We use the mean vectors μ as the final latent representations.

Final Prediction and Joint Model Training

The final stage predicts the association likelihood for any lncRNA-disease pair. A comprehensive feature vector is constructed by concatenating the latent representations (μ vectors) from the VGAE with the pre-computed nonlinear features from Phase 1. This vector is input to a specialized Multilayer Perceptron (MLP).

Each standard hidden layer in the MLP is defined as:

$$H_i = \text{ReLU}(WH_{i-1} + b) \tag{24}$$

where H_{i-1} is the output of the previous hidden layer, H_i is the output of the current hidden layer, and W and b are learnable weight and bias terms.

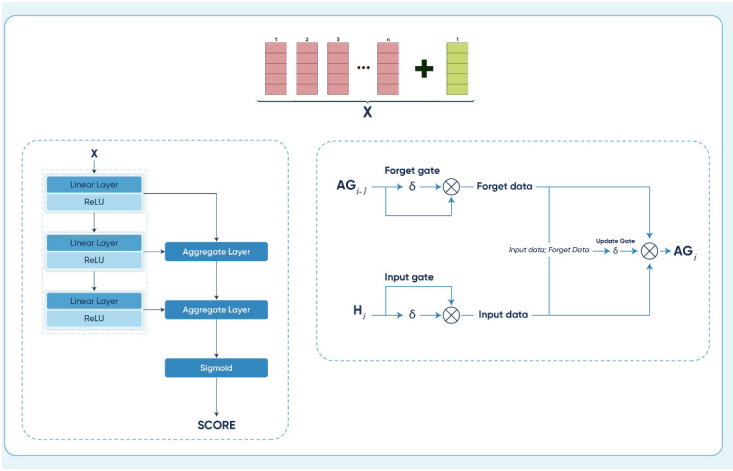


Fig. 5. Prediction network with gated aggregation. Concatenated features are processed by an MLP with alternating hidden and gated aggregation layers. These gates adaptively integrate information, and a final sigmoid layer outputs the association score.

To learn robust feature interactions, each standard layer (H_i) is paired with a novel aggregation layer (AG_i). As shown in Fig. 5, this aggregation layer uses three gates:

1. **Forget Gate**: Controls what information from the previous aggregation layer AG_{i-1} to retain.

$$FW_i = \text{Sigmoid}(AG_{i-1}W + b) \tag{25}$$

$$FD_i = FW_i \otimes AG_{i-1} \tag{26}$$

2. **Input Gate**: Selectively filters new information from the current hidden layer H_i.

$$IW_i = \text{Sigmoid}(H_iW + b) \tag{27}$$

$$ID_i = IW_i \otimes H_i \tag{28}$$

3. **Update Gate**: Integrates the filtered information from the forget gate (FD_i) and the input gate (ID_i).

$$UIW_i = \text{Sigmoid}([ID_i : FD_i]W + b) \tag{29}$$

$$AG_i = UIW_i \otimes ID_i + (1 - UIW_i) \otimes FD_i \tag{30}$$

where $[ID_i : FD_i]$ represents the concatenation operation.

After processing through all layers, a Sigmoid function yields a prediction score p between 0 and 1. This prediction head is optimized using the binary cross-entropy loss, $\mathcal{L}_{\text{link}}$.

$$\mathcal{L}_{\text{link}} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{31}$$

The entire model is trained end-to-end by optimizing a composite loss function, $\mathcal{L}_{\text{total}}$, which is a weighted sum of the reconstruction, KL divergence, and link prediction losses:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{kl}}\mathcal{L}_{\text{KL}} + \lambda_{\text{link}}\mathcal{L}_{\text{link}} \tag{32}$$

where λ_{recon}, λ_{kl} and λ_{link} are balancing hyperparameters.

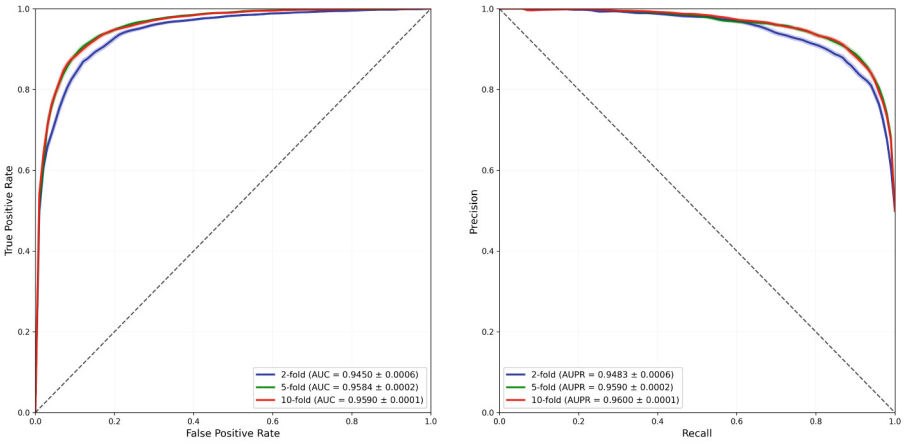
3 Experiment and Result

3.1 Experimental Setup

We report AUC and AUPR to account for class imbalance and data sparsity. We used k-fold cross-validation, masking positive lncRNA–disease associations in each test fold. Phase 1 (GCN-AE) follows LDAGM [21] (learning rate 10^{-2}, weight decay 10^{-5}, 128-d embedding, 100 epochs). Phase 2 uses learning rate 3×10^{-4}, weight decay 2×10^{-4}, and dropout 0.2.

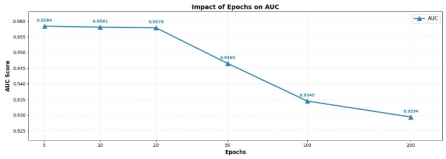
3.2 Results

Figure 6 reports ROC/PR curves under 2-, 5-, and 10-fold CV. The model achieves AUC/AUPR of $0.9450 \pm 0.0006/0.9483 \pm 0.0006$ (2-fold), $0.9584 \pm 0.0002/0.9590 \pm 0.0002$ (5-fold), and $0.9590 \pm 0.0001/0.9600 \pm 0.0001$ (10-fold). We use 5-fold CV in the remaining experiments as a balance between stability and efficiency. Figure 7 shows fast convergence, with performance peaking early and then saturating.



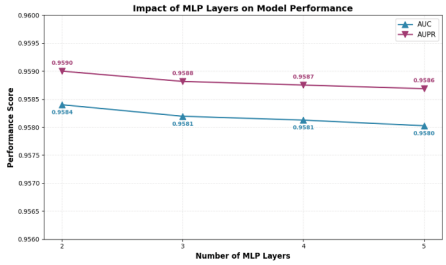
(a) The ROC Curve (b) The PR Curve

Fig. 6. Performance of the proposed model under 2-fold, 5-fold, and 10-fold cross-validation. (A) ROC curves (B) PR curves.

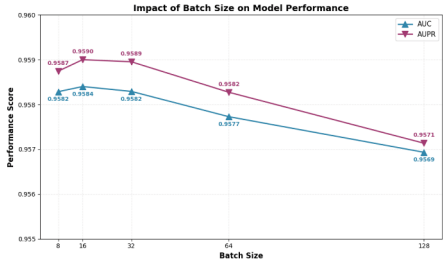


(a) Impact of epochs on AUC score (b) Impact of epochs on AUPR score

Fig. 7. Effect of training epochs on model performance: (A) AUC score, (B) AUPR score.



(a) MLP layers (b) MLP hidden dimension



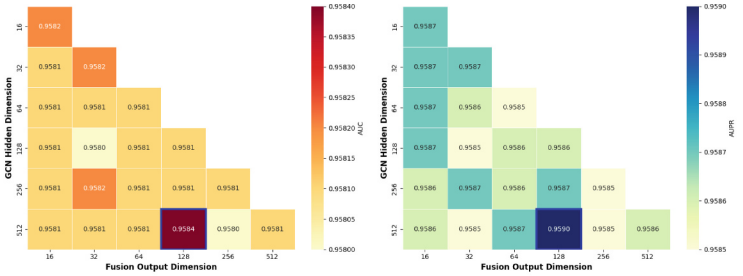
(c) Batch size (d) Network number

Fig. 8. Performance sensitivity across four hyperparameters: (A) MLP layers; (B) MLP hidden dimension; (C) batch size; (D) number of networks.

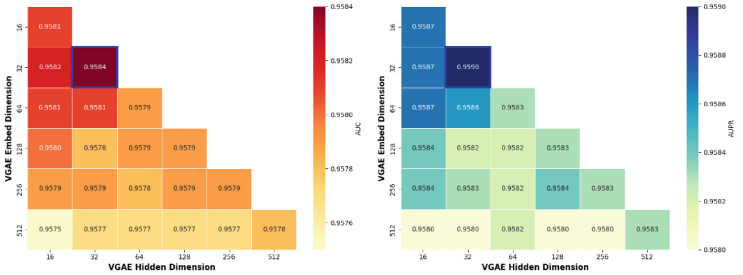
3.3 Hyperparameter Sensitivity Analysis

Figure 8 shows the model's sensitivity to four key hyperparameters: MLP layers (L), MLP hidden dimension (d_h), batch size (B), and the number of network layers (N). We tested $L \in \{2, 3, 4, 5\}$, $d_h \in \{10, 20, 30, 40, 50, 60\}$, $B \in \{8, 16, 32, 64, 128\}$ and $N \in \{1, 2, 3, 4, 5, 6\}$. Performance was robust across these ranges; we used L=2, d_h=50, B=16, and N=4.

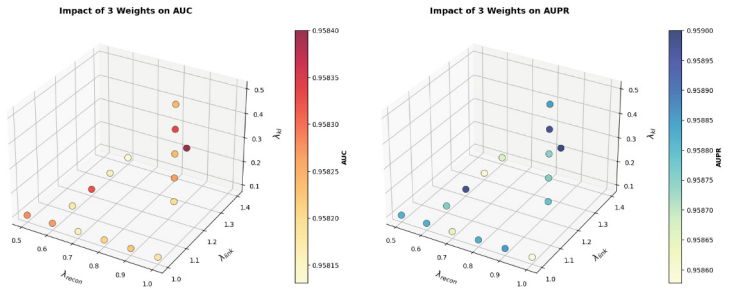
As illustrated in Fig. 9, Joint sweeps over encoder/fusion dimensions and loss weights identified the best setting (AUC=0.9584, AUPR=0.9590) with GCN hidden dimension 512, fusion output 128, VGAE hidden 32, VGAE embed 32, and $(\lambda_{\text{recon}}, \lambda_{\text{link}}, \lambda_{\text{kl}}) = (0.9, 1.3, 0.3)$.



(a) GCN hidden dimension (H) vs. Fusion output dimension (F): AUC (left) and AUPR (right).



(b) VGAE embed dimension (E) vs. VGAE hidden dimension (H): AUC (left) and AUPR (right).



(c) Effect of λ_{recon}, λ_{link}, and λ_{kl} on AUC (left) and AUPR (right); best region at high link weight and moderate KL.

Fig. 9. Hyperparameter sweeps over encoder and fusion dimensionalities, and effect of loss weights.

To assess robustness, we benchmarked against DBATNDA [4], HGC-GAN [13] and NAGTLDA [8]. As shown in Fig. 10, we evaluate all methods with AUC and AUPR. Our model attains the best performance on both metrics (AUC 0.9584, AUPR 0.9590). On AUC, it consistently outperforms the strongest baselines. On AUPR, the advantage is more pronounced: our model exceeds other

Table 1. Performance of our model on two datasets

Dataset	AUC	AUPR	MCC	ACC	Precision	Recall	F1-Score
Dataset 1	0.9584 ± 0.0002	0.9590 ± 0.0002	0.7873 ± 0.0006	0.8935 ± 0.0003	0.8951 ± 0.0020	0.8974 ± 0.0033	0.8939 ± 0.0007
Dataset 2	0.9300 ± 0.0003	0.9344 ± 0.0005	0.7093 ± 0.0030	0.8540 ± 0.0021	0.8739 ± 0.0086	0.8468 ± 0.0092	0.8526 ± 0.0024

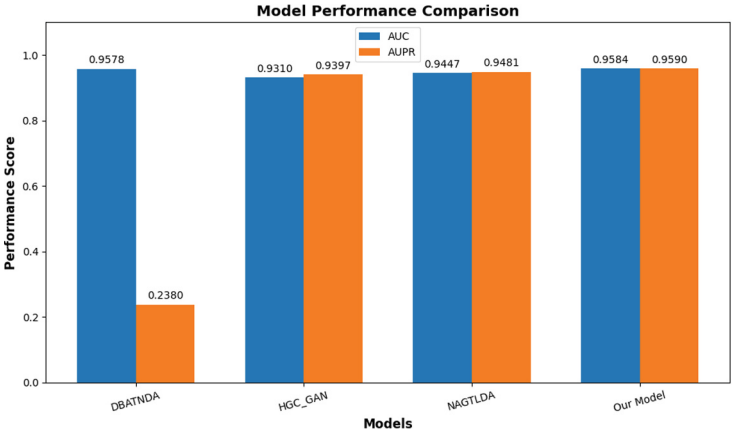


Fig. 10. Comparison of our model against other models on the AUC and AUPR metrics.

models by large margins. Furthermore, the detailed performance results of our proposed model on both datasets are described in Table 1.

4 Conclusion

In this study, we proposed an end-to-end deep learning framework for predicting lncRNA-disease associations that addresses key limitations of existing methods. Our principal contributions lie in a multi-stage, integrated architecture. First, we developed a hybrid feature learning strategy, uniquely combining pre-computed non-linear features with dynamically learned, task-specific embeddings. Second, we employ a multi-view GCN mechanism with attention-based fusion to automatically learn and weigh the importance of complementary biological information sources. Third, the framework incorporates a Variational Graph Autoencoder (VGAE) to learn probabilistic node embeddings, capturing structural uncertainty and providing regularization against overfitting in sparse networks. Finally, we used a specialized gated aggregation mechanism (a multilayer perceptron with forget/input/update gates) that selectively integrates information across layers, significantly improving the model's capacity to learn complex feature interactions. Comprehensive experiments demonstrate superior performance, achieving a state-of-the-art AUC of 0.9584 and AUPR of 0.9590 on Dataset 1, with the substantial improvement in AUPR being critical given the severe class imbalance in lncRNA–disease data.

While our model demonstrates strong predictive performance, we acknowledge several limitations. Performance is fundamentally dependent on input data quality, and the model's deep learning components remain "black-box" models, limiting biological interpretability. The framework also faces cold-start challenges for novel lncRNAs or diseases, and its generalizability to other organisms remains unexplored. Future work will focus on addressing these issues. We aim to explore efficient approximations or pruning strategies to reduce computational overhead and incorporate explainable AI (xAI) techniques to enhance model transparency. To address cold-start limitations, we plan to integrate alternative data modalities and develop transfer learning strategies. Broader directions include scaling the framework to larger biological networks and integrating diverse clinical data. Overall, while these limitations exist, our framework provides a strong foundation for future developments in computational biology.

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